

Functional MRI approach to developmental methylmercury and polychlorinated biphenyl neurotoxicity



Prenatal and early childhood exposure to methylmercury (MeHg) or polychlorinated biphenyls (PCBs) are associated with deficits in cognitive, sensory, motor and other functions measured by neurobehavioral tests. The main objective of this pilot study was to determine whether functional magnetic resonance imaging (fMRI) is effective for visualization of brain function alterations related to neurobehavior in subjects with high prenatal exposure to the two neurotoxicants, MeHg and PCBs. Twelve adolescents (all boys) from a Faroese birth cohort assembled in 1986–1987 were recruited based on their prenatal exposures to MeHg and PCB. All underwent fMRI scanning during behavioral tasks at age 15 years. Subjects with high mixed exposure to MeHg and PCBs were compared to those with low mixed exposure on fMRI photic stimulation and a motor task. Boys with low mixed exposures showed patterns of fMRI activation during visual and motor tasks that are typical of normal control subjects. However, those with high exposures showed activation in more areas of the brain and different and wider patterns of activation than the low mixed exposure group. The brain activation patterns observed in association with increased exposures to MeHg and PCBs are meaningful in regard to the known neurotoxicity of these substances. This methodology therefore has potential utility in visualizing structural neural system determinants of exposure-induced neurobehavioral dysfunction.